TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

Reminders

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product -1)

Outside $f(x) + k$: up/down · Inside $f(x - h)$: right h, $f(x + h)$: left h (opposite the sign)

Horizontal: $y = \#$, slope 0 · Vertical: $x = \#$, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

1. Kai graphed $f(x)$ and $g(x) = f(x + 5)$. Which statement describes how the graphs are related?

Tip: Inside the parentheses: move the opposite way of the sign.

A. translated right 5 units

B. translated up 5 units

C. translated left 5 units

2. A pool rental costs $50 plus $30 per hour, for at most 4 hours. What is the range for one rental?

> **Tip:** Range = the y-values (dollars). Find the cost at 0 hours and at 4 hours.

F. $50 \le y \le 170$

G. $0 \le x \le 4$

H. $30 \le y \le 170$

3. What is the slope of a line perpendicular to $y = -4/5x + 2$?

> **Tip:** Flip the fraction and change the sign.

A. $-5/4$

B. $5/4$

C. $4/5$

4. A line passes through (0, −3) and (6, 0). What is the zero of the function?

> **Tip:** The zero is the x-value where the line crosses the x-axis.

F. −3

G. 1/2

H. 6